SUMMARY · MK 6 BUSINESS ANALYTICS · WEEK 9

Operations Analytics: Bottlenecks, Production Optimisation & Stock Simulation

Analytics is used across operations: production planning, demand forecasting, inventory, predictive maintenance, quality control and delivery. This week uses one fictional bakery for two cases. Case 1 (optimisation): the head of production asks for a second packing line because packing used the most hours — but utilisation (hours used ÷ hours available) shows the oven is the bottleneck. Ranking products by margin per oven hour and using a solver lifts margin from IDR 855m to IDR 936m with the same machines; the shadow price shows one extra oven hour is worth IDR 1.5m, while a second packing line is worth IDR 0. Case 2 (Monte Carlo simulation): setting the flour reorder point when daily use and delivery times vary; the old rule (64 tonnes) runs out on 47.3% of orders, while 96 tonnes gives the lowest total cost, IDR 15.4m a month.

Material

1 Where analytics is used in operations

Production planning
What should the factory make this month? *(case 1)*

Demand forecasting
How much will customers order? (see weeks 4 & 7)

Inventory
When should we reorder stock? *(case 2)*

Predictive maintenance
Which machine will fail next?

Quality control
Is this batch still within limits? (see MK4 week 9)

Delivery
When will the order reach the customer?

Case 1 · Optimisation	Case 2 · Simulation

Situation	Monthly production at a bakery	Flour stock at the same bakery
Decision	How much of each product to make	When to reorder
Method	Optimisation (linear programming + solver)	Monte Carlo simulation

Both are **prescriptive analytics** (week 5): not just describing or forecasting, but recommending a decision. The company and all figures are fictional.

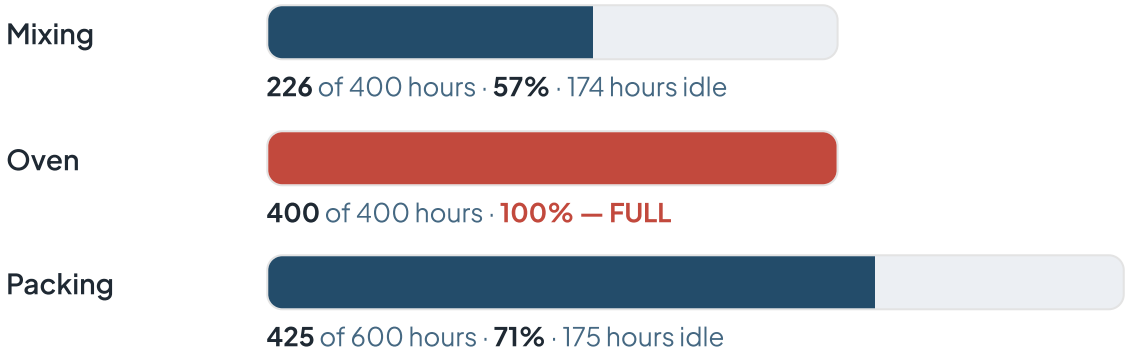
2 Case 1: a request for a second packing line

A bakery makes three products: **bread, biscuits and cakes**. Each month it decides how many packs of each to make; the goal is the highest total monthly margin. Every pack passes three stations (25 working days a month):

Station	Shifts	Hours available/month
Mixing	2 shifts	400 hours
Oven	2 shifts	400 hours
Packing	3 shifts	600 hours

Last month packing ran **425 hours** — more than any other station. The head of production asks for a **second packing line**. The decision: approve it or not?

Hours used vs hours available last month



Grey bar length = hours available; coloured part = hours used. Packing has the longest filled bar (425 hours) but still 175 hours spare. The oven is completely full — that is the bottleneck.

```
Utilisation = Hours used ÷ Hours available
```

Mixing 226 ÷ 400 = **57%**; oven 400 ÷ 400 = **100%**; packing 425 ÷ 600 = **71%**. **The oven is full.** Hours used do not show the bottleneck; hours used out of hours available do. Packing only looks "busiest" because it has more shifts.

> ! **First lesson**
> Before buying capacity, find the station that really limits output. Adding capacity at a non-bottleneck adds no output at all (recall the theory of constraints in MK4).

3 The optimisation problem and margin per oven hour

Element	Content
Objective	Highest total margin for the month
Levers (decision variables)	Thousand packs of bread, biscuits, cakes
Constraints	Hours available at each station; demand for each product
Method	A solver finds the highest-margin plan that meets every constraint

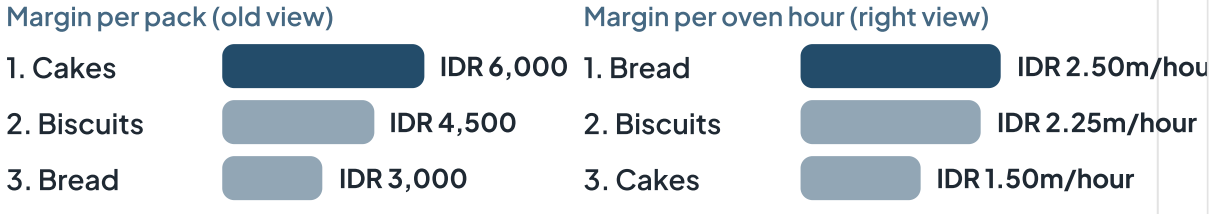
	Bread	Biscuits	Cakes
Margin per pack	IDR 3,000	IDR 4,500	IDR 6,000
Mixing hours per 1,000 packs	1.0	0.8	1.2
Oven hours per 1,000 packs	1.2	2.0	4.0
Packing hours per 1,000 packs	1.5	2.5	2.5
Demand (thousand packs/month)	180	80	60
Last month's plan (thousand packs)	150	50	30

```
Max 3R + 4.5B + 6K (IDR million; R, B, K in thousand packs)
subject to: 1.0R + 0.8B + 1.2K ≤ 400 (mixing) · 1.2R + 2.0B + 4.0K
≤ 400 (oven) · 1.5R + 2.5B + 2.5K ≤ 600 (packing) · R ≤ 180, B ≤
80, K ≤ 60, all ≥ 0
```

Last month's plan favoured cakes because they have the highest margin per pack. But when the oven is full, the right measure is **margin per oven hour**:

```
Margin per bottleneck hour = Margin per 1,000 packs ÷ Oven hours
per 1,000 packs
```

The product ranking flips when the oven is full



The right side is margin per 1,000 packs ÷ oven hours per 1,000 packs (bread: IDR 3m ÷ 1.2 hours = IDR 2.5m/hour). Cakes win on the left but come last on the right.

Bread IDR 3.0m ÷ 1.2 = **IDR 2.50m/hour**; biscuits 4.5 ÷ 2.0 = **IDR 2.25m/hour**; cakes 6.0 ÷ 4.0 = **IDR 1.50m/hour**. Cakes earn the most per pack but *the least per oven hour*.

Rule of thumb

When one resource is full, rank products by margin per unit of that scarce resource — not by margin per unit of product.

4 The optimal plan and the value of one extra hour (shadow price)

Thousand packs/month	Last month	Optimal	Demand
Bread	150	**180**	180
Biscuits	50	**80**	80

Cakes	30	**6**	60
Total margin	IDR 855m	**IDR 936m**	

How to read it: bread and biscuits are made up to demand (most valuable per oven hour). Oven hours: 180 × 1.2 + 80 × 2.0 = 216 + 160 = 376; the remaining 24 hours go to cakes = 24 ÷ 4 = 6 thousand packs. Margin = 180 × 3 + 80 × 4.5 + 6 × 6 = 540 + 360 + 36 = **IDR 936m** — up **IDR 81m a month with the same machines**. Packing check: 270 + 200 + 15 = 485 hours ≤ 600 ✓.

Shadow price = the extra margin from one more hour at a station, under the optimal plan:

Mixing
IDR 0 — hours still idle

Oven
IDR 1.5m — one more hour = 0.25 thousand packs of cake × IDR 6m

Packing
IDR 0 — hours still idle

The station with a positive shadow price is the bottleneck. An extra hour has value only at the full station.

Where the 400 oven hours go

Last month → margin IDR 855m



- Bread: **150k packs** × 1.2 oven h per 1k packs = **180 h**
- Biscuits: **50k packs** × 2 oven h per 1k packs = **100 h**
- Cakes: **30k packs** × 4 oven h per 1k packs = **120 h**

Optimal plan → margin IDR 936m (+IDR 81m)



- Bread: **180k packs** × 1.2 oven h per 1k packs = **216 h**
- Biscuits: **80k packs** × 2 oven h per 1k packs = **160 h**
- Cakes: **6k packs** × 4 oven h per 1k packs = **24 h**

Coloured blocks show OVEN HOURS only (400 hours in total). Product volume (thousand packs) is in the calculation line under each bar: packs × oven hours per 1,000 packs = oven hours. The oven is full both times (400 hours); the difference is moving oven hours from cakes (IDR 1.5m/hour) to bread and biscuits (IDR 2.5m and 2.25m/hour).

Plan	Oven hours	Packing hours	Margin/month
Last month's plan	400	600	IDR 855m
Optimal	400	600	IDR 936m
Optimal + second packing line	400	1,200	IDR 936m
Optimal + third oven shift	600	600	IDR 1,224m
Optimal + both	600	1,200	IDR 1,236m

Each extra oven hour is worth IDR 1.5m up to about 584 hours; after that **packing is full and becomes the new bottleneck**. With 600 oven hours the solver cuts biscuits to 72 thousand packs so cakes can reach 60 thousand, for a margin of IDR 1,224m. The second packing line is worth **IDR 0 now**, and only **IDR 12m** after a third oven shift.

≡ **Recommendation**

Reject the second packing line. Change the production plan first (+IDR 81m, no capital cost), then evaluate a third oven shift: worthwhile if its extra cost is below about IDR 288m/month (IDR 1,224m – 936m).

5 Case 2: the flour reorder point when use and delivery time vary

The same bakery uses about **8 tonnes of flour per working day** and orders **100 tonnes** at a time (about twice a month). The supplier takes **5–12 days** to deliver. Current rule: reorder when stock hits **64 tonnes** (8 tonnes/day × 8 days average delivery).

Stock falls → Hits the reorder point → order → Wait for delivery (lead time) → Delivery arrives, stock rises again

Stock runs out if the flour used while waiting is more than the reorder point.

Historical data	Average	Range
Daily use (last 300 working days)	8.0 tonnes	4.0 – 11.8 tonnes
Delivery time (last 40 deliveries)	8 days	5 – 12 days

Monte Carlo simulation (no distribution assumed — every draw comes from the bakery's own records):

1. Draw a delivery time Pick one of the last 40 deliveries at random.

2. Draw daily use Pick that many days at random from the last 300.

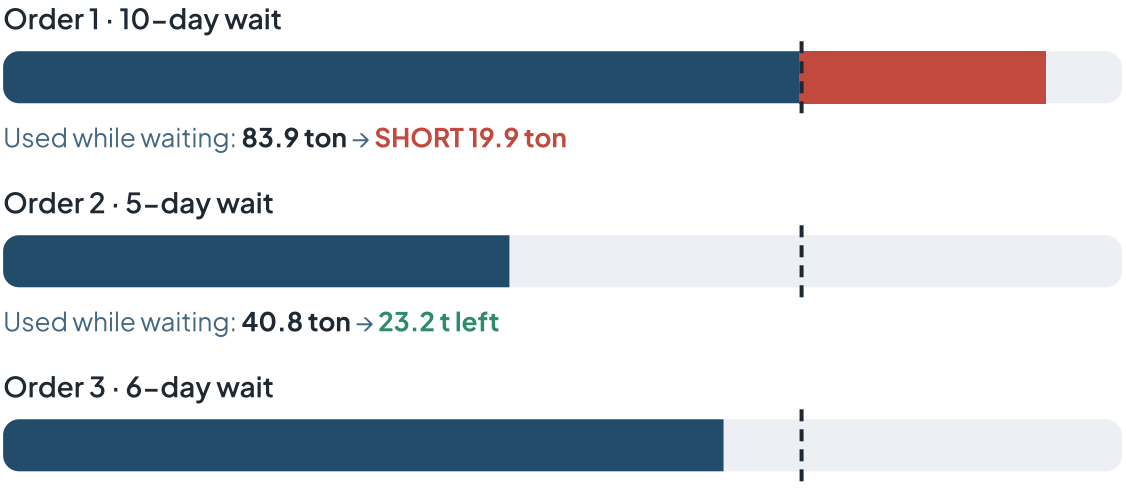
3. Add up Total flour used while waiting.

4. Repeat10,000 orders.

Order	Delivery time	Daily use drawn (tonnes)	Used	Reorder at 64
1	10 days	8.7; 7.0; 8.4; 11.2; 6.8; 8.4; 9.9; 8.1; 7.3; 8.1	83.9	**19.9 short**
2	5 days	9.0; 8.1; 6.1; 10.5; 7.1	40.8	23.2 left
3	6 days	11.6; 10.3; 10.3; 8.3; 7.1; 10.3	57.9	6.1 left

One of three runs out. Over 10,000 simulations: average used while waiting **64.5 tonnes**, and **47.3%** of orders run out under the current rule.

Three simulated orders: is 64 tonnes of stock enough?



Used while waiting: **57.9 ton → 6.1 t left**

Line = 64-tonne reorder point (stock when ordering)

The red part = flour needed but no longer in stock (a stock-out). Of 10,000 simulated orders, 47.3% end up like order 1.

Why the average rule fails

A reorder point equal to the average use while waiting runs out on *about half* of all orders — because half the time, high use and slow delivery together exceed the average. Averages hide variation; simulation shows it.

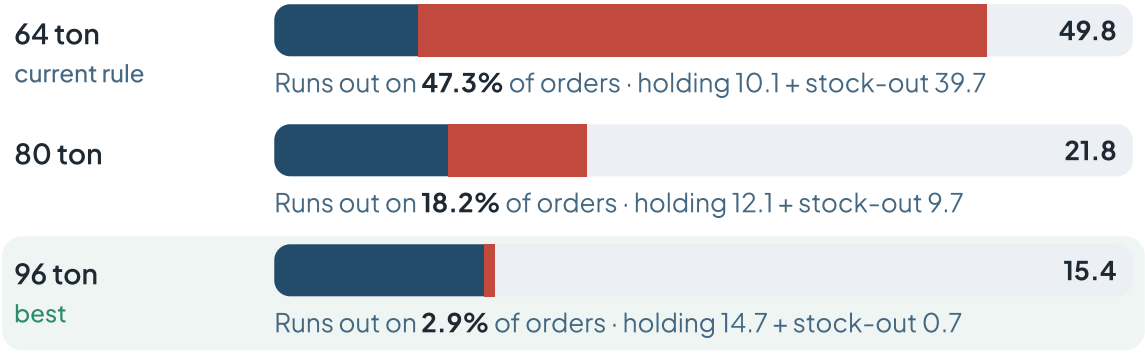
6 Holding cost vs stock-out cost, and the best reorder point

Component	Assumption
Holding cost	1 tonne in stock = IDR 180,000/month (2% of the IDR 9m price)
Stock-out cost	1 tonne short stops the oven for 2 hours (the oven uses 8 tonnes in a 16-hour day)
Value of an oven hour	IDR 1.5m — *the shadow price from case 1*

```
Cost per tonne short = 2 hours × IDR 1.5m = IDR 3m
```

Notice how the two cases connect: the cost of running out of flour comes from the value of the bottleneck hour in the optimisation case. Running out is expensive precisely because the oven is the bottleneck.

Total monthly cost for each reorder point





Raising the reorder point adds a little holding cost (blue) but cuts stock-out cost fast (red). The lowest total is at 96 tonnes: IDR 15.4m, saving IDR 34.5m versus the current rule.

Reorder point	Orders that run out	Holding	Running out	Total (IDR m/month)
64 tonnes (current rule)	47.3%	10.1	39.7	49.8
80 tonnes	18.2%	12.1	9.7	21.8
96 tonnes	2.9%	14.7	0.7	**15.4**
112 tonnes	0.0%	17.6	0.0	17.6

Below 96 tonnes the stock-out cost rises fast; above it, holding cost slowly takes over. **Reordering at 96 tonnes saves IDR 34.5m a month** versus the current rule. The gap 96 – 64.5 ≈ 31.5 tonnes is the **safety stock**, justified by cost rather than gut feel.

≡ **Slide exercise: a cheaper supplier**

New supplier: IDR 8.98m/tonne (vs 9.00m), but delivery 4–18 days (average 9.2 days vs 8.0). Price saving about IDR 4m/month (about 200 tonnes × IDR 20,000). *How to answer*: simulate both suppliers with their own delivery data, find the best reorder point for each, then compare total monthly cost (price + holding + stock-out). Longer, more variable delivery needs much more safety stock; with a shortage cost of IDR 3m/tonne, the extra holding and stock-out cost can easily exceed the IDR 4m saving. Decide with simulation numbers, not price per tonne alone.

7 Optimisation vs simulation: when to use which

Optimisation	Monte Carlo simulation

Question	"What is the best choice under these constraints?"	"How often does something bad happen under this rule?"
Inputs	Fixed numbers: margins, hours per unit, capacity, demand	Random variables: drawn from historical data or distributions
Output	One optimal plan + shadow prices	A distribution of outcomes (e.g. % of orders running out)
Weakness	Ignores uncertainty if inputs are treated as certain	Does not directly give "the best" — you compare scenarios
This week's example	Bread/biscuit/cake production plan	Flour reorder point

They are often used together: simulation gives each option's cost or risk, then we pick the cheapest option (as in comparing reorder points 64/80/96/112).

Common mistakes these two cases avoid

(1) Assuming the station with the most hours used is the bottleneck. (2) Ranking products by margin per unit instead of per bottleneck hour. (3) Buying capacity at a station that is not full. (4) Setting the reorder point = average use while waiting. (5) Comparing suppliers on unit price alone.

Key takeaways

1. Operations analytics: production planning, forecasting, inventory, predictive maintenance, quality control, delivery.
2. The bottleneck is found from utilisation (hours used ÷ hours available), not hours used alone.
3. When one resource is full, rank products by margin per hour of that resource.
4. The bakery's optimal plan: bread 180, biscuits 80, cakes 6 (thousand packs) → margin IDR 936m vs 855m.
5. Shadow price = the value of one extra hour; positive only at the full station (oven IDR 1.5m/hour; mixing & packing IDR 0).
6. A second packing line is worth IDR 0 now; a third oven shift lifts margin to IDR 1,224m until packing becomes the new bottleneck.
7. Monte Carlo: draw delivery times and daily use at random from your own data, repeat 10,000 times.
8. Reorder point = average use while waiting → runs out on about half of orders (64 tonnes: 47.3%).
9. Cost of 1 tonne short = 2 oven hours × IDR 1.5m = IDR 3m; the best reorder point is 96 tonnes (IDR 15.4m/month, saving 34.5m).
10. Compare suppliers with a simulated total cost, not unit price.

Case studies

Worked examples

Problem 1 · Easy **Utilisation and the bottleneck**

A packaging plant: cutting used 300 of 400 hours; printing 360 of 360; folding 450 of 720. Which station is the bottleneck? Does adding a folding machine make sense?

1. `Utilisation = Hours used ÷ Hours available`
2. Cutting 300 ÷ 400 = 75%; printing 360 ÷ 360 = **100%**; folding 450 ÷ 720 = 62.5%.
3. Bottleneck = **printing**. Folding uses the most hours but is still 37.5% idle, so another folding machine adds no output.

Answer: Bottleneck: printing (100%). Don't add a folding machine.

Problem 2 Medium **Filling the bottleneck by margin per hour**

The oven has 400 hours. Bread: margin IDR 3m/1,000 packs, 1.2 oven hours, demand 180. Biscuits: IDR 4.5m, 2.0 hours, demand 80. Cakes: IDR 6m, 4.0 hours, demand 60. Build the margin-maximising plan (assume other stations don't bind).

1. Margin per oven hour: bread 3 ÷ 1.2 = 2.50; biscuits 4.5 ÷ 2 = 2.25; cakes 6 ÷ 4 = 1.50 (IDR m). Order: bread → biscuits → cakes.
2. Bread to demand: 180 × 1.2 = 216 hours (184 left).
3. Biscuits to demand: 80 × 2 = 160 hours (24 left).
4. Cakes: 24 ÷ 4 = 6 thousand packs.
5. Margin = 180 × 3 + 80 × 4.5 + 6 × 6 = 540 + 360 + 36 = **IDR 936m**.

Answer: Bread 180, biscuits 80, cakes 6 (thousand packs); margin IDR 936m.

Problem 3 Medium **Using the shadow price for a capacity decision**

The oven's shadow price is IDR 1.5m/hour (valid up to about 584 hours). A third oven shift adds 200 hours at IDR 220m/month. Worth it? What is the most the shift could cost?

1. Hours worth IDR 1.5m: from 400 to 584 = 184 hours → 184 × 1.5 = IDR 276m.
2. The last 16 hours are worth less because packing is already full; the solver gives total margin IDR 1,224m at 600 hours → total gain = 1,224 – 936 = **IDR 288m**.
3. IDR 288m > IDR 220m → **worth it**, about IDR 68m extra profit a month. The most it could cost and still pay is about IDR 288m/month.

Answer: Worth it (extra margin IDR 288m vs cost IDR 220m).

Problem 4 Challenging **Total cost of a reorder point**

(Illustrative numbers.) The simulation gives: at a reorder point of 88 tonnes, holding cost is IDR 13.4m/month and average shortage is 1.2 tonnes a month. Shortage cost is IDR 3m/tonne. Compare with 96 tonnes (total IDR 15.4m).

1. ```
   Total cost = Holding cost + Shortage (tonnes) × Cost per tonne short
   ```

2. Stock-out = 1.2 × 3 = IDR 3.6m. Total at 88 t = 13.4 + 3.6 = **IDR 17.0m**.

3. 17.0 > 15.4 → 96 t is still about IDR 1.6m/month cheaper. The high shortage cost (the oven is the bottleneck) makes a larger safety stock worthwhile.

Answer: 88 t: IDR 17.0m/month; 96 t remains best.

Practice questions

Try answering first. The answer key is below each question.

1. Packing used 425 hours (of 600), the oven 400 (of 400). The bottleneck is…

A. Packing, most hours
B. The oven, 100% utilised
C. Mixing
D. None

Answer: B. The bottleneck is hours used ÷ hours available.

2. When the oven is full, products should be ranked by…

A. Margin per pack
B. Selling price
C. Margin per oven hour
D. Demand

Answer: C. The scarce resource sets the right measure.
   ```

3. Cakes earn IDR 6m per 1,000 packs and need 4 oven hours. Margin per oven hour is…

A. IDR 1.5m
B. IDR 2.25m
C. IDR 2.5m
D. IDR 24m

Answer: A. 6 ÷ 4 = 1.5 — the lowest of the three.

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4. The optimal plan lifts margin from IDR 855m to…

A. IDR 900m
B. IDR 936m
C. IDR 1,224m
D. IDR 1,236m

Answer: B. Bread 180 + biscuits 80 to demand, remaining oven to cakes 6 → IDR 936m.

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5. Packing's shadow price under the optimal plan is IDR 0 because…

A. Packing is cheap
B. Packing still has idle hours
C. Packing doesn't matter
D. The solver is wrong

Answer: B. Extra hours only have value at a full station.

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6. The value of one extra oven hour under the optimal plan is…

A. IDR 0
B. IDR 1.5m
C. IDR 2.5m
D. IDR 6m

Answer: B. One hour = 0.25 thousand packs of cake × IDR 6m = IDR 1.5m.

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7. A second packing line (without more oven) adds…

A. IDR 0
B. IDR 12m
C. IDR 81m
D. IDR 288m

Answer: A. While the oven is the bottleneck, extra packing capacity goes unused.

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8. After a third oven shift (600 hours), the next bottleneck is…

A. Mixing
B. Packing
C. The oven again
D. Bread demand

Answer: B. Packing fills at about 584 oven hours; after that its shadow price turns positive.

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9. In the flour Monte Carlo simulation, the daily-use distribution is…

A. Assumed normal
B. Drawn directly from the bakery's 300 days of data
C. Assumed fixed at 8 tonnes
D. Taken from the supplier

Answer: B. No distribution is assumed; every draw comes from the bakery's records.

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10. Delivery 10 days, total use drawn 83.9 tonnes, reorder point 64. The result is…

A. 19.9 tonnes left
B. 19.9 tonnes short
C. Exactly enough
D. 83.9 tonnes short

Answer: B. 83.9 − 64 = 19.9 tonnes short.

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11. A 64-tonne reorder point (= average use while waiting) runs out on…

A. 2.9% of orders
B. 18.2% of orders
C. 47.3% of orders
D. 0% of orders

Answer: C. Reordering at the average runs out about half the time.

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12. The cost of being 1 tonne short is IDR 3m because…

A. Flour prices rise
B. 1 tonne = 2 lost oven hours × IDR 1.5m per oven hour
C. A supplier penalty
D. Warehouse cost

Answer: B. The oven-hour value comes from case 1's shadow price.

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13. The reorder point with the lowest total monthly cost is…

A. 64 t
B. 80 t
C. 96 t
D. 112 t

Answer: C. 96 t: IDR 15.4m (holding 14.7 + stock-out 0.7).

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14. Why is 112 t dearer than 96 t even though it never runs out?

A. Holding cost rises more than the stock-out saving
B. The supplier raises prices
C. The oven slows
D. The simulation is wrong

Answer: A. Holding 17.6 vs (14.7 + 0.7) = 15.4.

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15. A new supplier is 0.2% cheaper but delivers in 4–18 days. The right way to decide is…

A. Pick the cheaper one
B. Simulate both and compare total monthly cost
C. Pick the faster average only
D. Split orders 50/50

Answer: B. Delivery variation raises safety stock and stock-out risk; compare it with the ~IDR 4m saving.

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16. The main difference between optimisation and simulation is…

A. Optimisation finds the best choice under constraints; simulation measures how often bad outcomes occur under uncertainty
B. Simulation is always more accurate
C. Optimisation is only for inventory
D. No difference

Answer: A. They are often combined: simulate each option's cost, then pick the cheapest.

Sources

- P3MD Week 9 MK6 — Operations Analytics (fictional bakery cases)